CLYC Full-Energy Peak Efficiency versus Source Distance

Simulation and Measurement Validation for the DCS-Monitor

DCS-Monitor simulation notes

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Abstract

The DCS-Monitor uses a CLYC scintillator to detect gamma radiation from a spent-fuel cask. This report compares the simulated full-energy peak efficiency of the CLYC detector with a nine-point measurement at source-to-detector distances of 200 mm, 300 mm, and 400 mm and gamma energies of 661.64 keV, 1173.23 keV, and 1332.51 keV. The simulation uses 1×10^7 incident gammas per point, an isotropic point source at the centre of a 2 mm-thick epoxy source puck, and the full CLYC detector and casing geometry. The measured efficiencies are taken from `effDist_clyc1.csv`; the column labelled `Efficiency (%)` is a factor 1000 larger than the value obtained from the measurement formula, so the corrected measured efficiency is that column divided by 1000.

At 661.64 keV, the simulation and measurement agree within 9 %: the simulation-to-measurement ratios are 1.059, 0.963, and 0.915 at 200 mm, 300 mm, and 400 mm. At 1173.23 keV, the simulation is 1.48–1.70 times the measurement, a discrepancy of 4–8 standard deviations using the simulation statistical error. At 1332.51 keV, the ratios are 1.353, 1.213, and 1.014; the first two points are 4.7 and 2.2 standard deviations high, while the 400 mm point is consistent with the measurement but has an 11.9 % statistical uncertainty. The measured 1173.23 keV/1332.51 keV ratio is 0.94, 0.92, and 0.90, whereas the reference column of the CSV gives 1.21 and the simulation gives 1.18, 1.12, and 1.49. The measured ratio is therefore inverted relative to both the reference and the simulation. A no-puck control at 200 mm and 661.64 keV gives 0.008 76(30) % versus 0.008 43(29) % with the puck, so the epoxy puck changes the efficiency by 3.8(48) %, consistent with zero. The 661.64 keV agreement validates the geometry, the source distance, and the analysis chain at the 10 % level. The energy-dependent discrepancy at 1173.23 keV and 1332.51 keV is not explained by the source puck; the most plausible remaining cause is that the simulation uses one monoenergetic gamma per event and does not model the 1173.23 —-1332.51 keV cascade of ^{60}Co or the detector response. A cascade source, a detector response model, and the measured spectra are the next validation steps.

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1 Motivation and scope

The DCS-Monitor simulation must predict the CLYC count rate for a source at a known distance. A full-energy peak efficiency that is wrong by a factor of 1.5 would change the inferred activity or the alarm threshold by the same factor. The measurement in `effDist_clyc1.csv` provides nine independent points: three distances and three gamma lines. This report uses those points to test the detector geometry, the source placement, and the analysis chain.

The scope is deliberately narrow. The simulation is a Geant4 transport of monoenergetic gammas through the CLYC detector and its casing. It does not simulate scintillation light, light collection, or the electronics. The efficiency is defined as the number of events whose total energy deposition in the crystal lies within $\pm 2\,\%$ of the incident gamma energy, divided by the number of incident gammas. The measurement is the full-energy peak efficiency obtained from the net peak area, the source activity, the emission probability, and the measurement time. The two definitions are compared at the same nominal source-to-detector distance.

The report is a validation note, not a final calibration. The statistical uncertainties are reported; the systematic uncertainties of the source activity, the distance, the detector geometry, and the peak fitting are not yet included. The conclusions are therefore statements about the consistency of the present simulation and data, not a complete uncertainty budget.

2 Experimental data

The measurement file is `/home/turko46/data/CLYC_validation/effDist_clyc1.csv`. It contains nine rows: ^{137}Cs at 661.64 keV and ^{60}Co at 1173.23 keV and 1332.51 keV, each at 200 mm, 300 mm, and 400 mm. The detector is labelled CLYC1. The initial activities are 4.77×10^6 Bq for ^{137}Cs and 18.50×10^6 Bq for ^{60}Co; the current activities at the measurement dates are 3.65×10^6 Bq and 11.31×10^6 Bq, respectively. The measurement time is 600 s for every row.

The experimental efficiency is

$$\epsilon_{\mathrm{exp}} = \frac{N_{\mathrm{peak}}}{A(t)\, p_\gamma\, t} \times 100, \tag{1}$$

where N_{peak} is the net peak area, $A(t)$ is the current activity in becquerel, $p_\gamma = P_\gamma/100$ is the emission probability per decay, and t is the measurement time in seconds. For the 200 mm 661.64 keV row,

$$\epsilon_{\mathrm{exp}} = \frac{148423}{(3.65 \times 10^6)(0.8510)(600)} \times 100 = 0.007\,96\,\%. \tag{2}$$

The CSV column labelled `Efficiency (%)` contains 7.96 for this row, which is exactly 1000 times the result of Eq. (2). The same factor of 1000 applies to all nine rows. The corrected measured efficiency used in this report is therefore

$$\epsilon_{\mathrm{meas}} = \frac{\texttt{Efficiency (\%)}}{1000}. \tag{3}$$

The origin of the factor is not documented in the CSV. It is not caused by the emission probability: using P_γ as a percentage instead of a fraction would change the result by a factor of 100, not 1000. The factor is consistent across all rows, so it is treated as a column-normalisation error and corrected once.

The CSV also contains an `Efficiency Reference Value (%)` column with values 6.64, 3.14, and 2.60 for the three energies. This column is not the measured efficiency and is not used as the denominator in this report. It is used only to compare the energy dependence: its 1173.23 keV/1332.51 keV ratio is 1.21. The measured ratio is 0.94, 0.92, and 0.90; the simulation ratio is 1.18, 1.12, and 1.49. The reference column therefore follows the simulation better than the measurement, which is one of the central results of this report.

Table 1: Experimental data from `effDist_clyc1.csv`. The column $\texttt{eff}_{\mathrm{CSV}}$ is the CSV value; the column $\texttt{eff}_{\mathrm{meas}}$ is that value divided by 1000, following Eq. (3). The current activity $A(t)$ is used in Eq. (1).

d [mm]	Nuclide	E [keV]	$A(t)$ [Bq]	P_γ [%]	t [s]	N_{peak}	$\mathrm{eff}_{\mathrm{CSV}}$ [%]	$\mathrm{eff}_{\mathrm{meas}}$ [%]
200	^{137}Cs	661.64	3.65×10^6	85.10	600	148423	7.96	0.00796
200	^{60}Co	1173.23	11.31×10^6	99.97	600	153103	2.26	0.00226
200	^{60}Co	1332.51	11.31×10^6	99.99	600	163177	2.41	0.00241
300	^{137}Cs	661.64	3.65×10^6	85.10	600	74617	4.00	0.00400
300	^{60}Co	1173.23	11.31×10^6	99.97	600	76163	1.12	0.00112
300	^{60}Co	1332.51	11.31×10^6	99.99	600	82693	1.22	0.00122
400	^{137}Cs	661.64	3.65×10^6	85.10	600	41666	2.24	0.00224
400	^{60}Co	1173.23	11.31×10^6	99.97	600	42612	0.63	0.00063
400	^{60}Co	1332.51	11.31×10^6	99.99	600	47252	0.70	0.00070

3 Geometry and source model

3.1 CLYC detector and source puck

The detector is a CLYC scintillator with a 25 mm diameter and a 25 mm length. The crystal front face is 2 mm behind the detector front plane. The crystal is mounted in the detector body and magnesium shield described by `GeometryCLYC.cc`; the body is included in the simulation, so the gamma must traverse the detector entrance window and the surrounding material before reaching the crystal. The crystal density is $3.31\,\mathrm{g\,cm^{-3}}$.

The source is an epoxy puck with a radius of 10 mm, a thickness of 2 mm, and a density of $1.16\,\mathrm{g\,cm^{-3}}$. The gamma source is at the puck centre. The puck is centred on the detector axis at $(0, 0, d)$, where d is the nominal source-to-detector distance. Because the crystal front face is 2 mm behind the detector front plane, the crystal-to-source distance is $d + 2$ mm. The source is isotropic in the production runs. The event-display macros described in Section 3.3 use a focused source only to make the tracks visible; they are not the production source model.

3.2 Geometric acceptance

For a point source on the detector axis and a circular crystal face of radius r, the fraction of the full solid angle subtended by the face is

$$\epsilon_\Omega(d) = \frac{1}{2}\left(1 - \frac{d}{\sqrt{d^2 + r^2}}\right), \tag{4}$$

where d is the crystal-to-source distance and $r = 12.5\,\mathrm{mm}$. Using $d = d_{\mathrm{nom}} + 2\,\mathrm{mm}$ gives

$$\epsilon_\Omega = \begin{cases} 0.0955\,\%, & d_{\mathrm{nom}} = 200\,\mathrm{mm}, \\ 0.0428\,\%, & d_{\mathrm{nom}} = 300\,\mathrm{mm}, \\ 0.0242\,\%, & d_{\mathrm{nom}} = 400\,\mathrm{mm}. \end{cases} \tag{5}$$

The geometric acceptance is not the detection efficiency. It is the fraction of emitted gammas that intersect the crystal front face. The intrinsic efficiency is the ratio of the full-energy peak efficiency to ϵ_Ω; it removes the leading $1/d^2$ dependence and is useful for comparing the three distances.

Table 2: Reference detector and source geometry. The crystal-to-source distance is the nominal distance plus 2 mm because the crystal front face is recessed behind the detector front plane. The geometric acceptance follows Eq. (4).

Quantity	Value	Note
Crystal material	CLYC	$3.31\,\mathrm{g\,cm^{-3}}$
Crystal dimensions	25 mm × 25 mm	diameter × length
Crystal front face	2 mm behind detector front	recessed
Detector body	29 mm snout	magnesium shield behind
Source puck material	epoxy	$1.16\,\mathrm{g\,cm^{-3}}$
Source puck dimensions	10 mm radius, 2 mm thick	disc
Source position	puck centre	$(0, 0, d)$
Nominal distances	200 mm, 300 mm, 400 mm	source to detector front
Crystal-to-source distance	$d + 2\,\mathrm{mm}$	used in Eq. (4)
Geometric acceptance	0.0955, 0.0428, 0.0242 %	at 200 mm, 300 mm, 400 mm

3.3 Visual check of the detector and source

Figure 1 shows two Geant4/OpenGL views of the detector, the source puck, and ten 661.64 keV gamma tracks. The source is at 200 mm. The top panel is a side view along the detector axis; the bottom panel is an oblique view. The detector body is drawn semi-transparently so that the crystal and the tracks inside the body are visible. The orange disc is the source puck. The tracks are green; several leave the detector body, which is expected for an isotropic source and a finite crystal. The source in these renders is focused toward the detector to keep the ten tracks in the field of view; the production runs use an isotropic source, so the renders are illustrative and are not used to compute the efficiency.

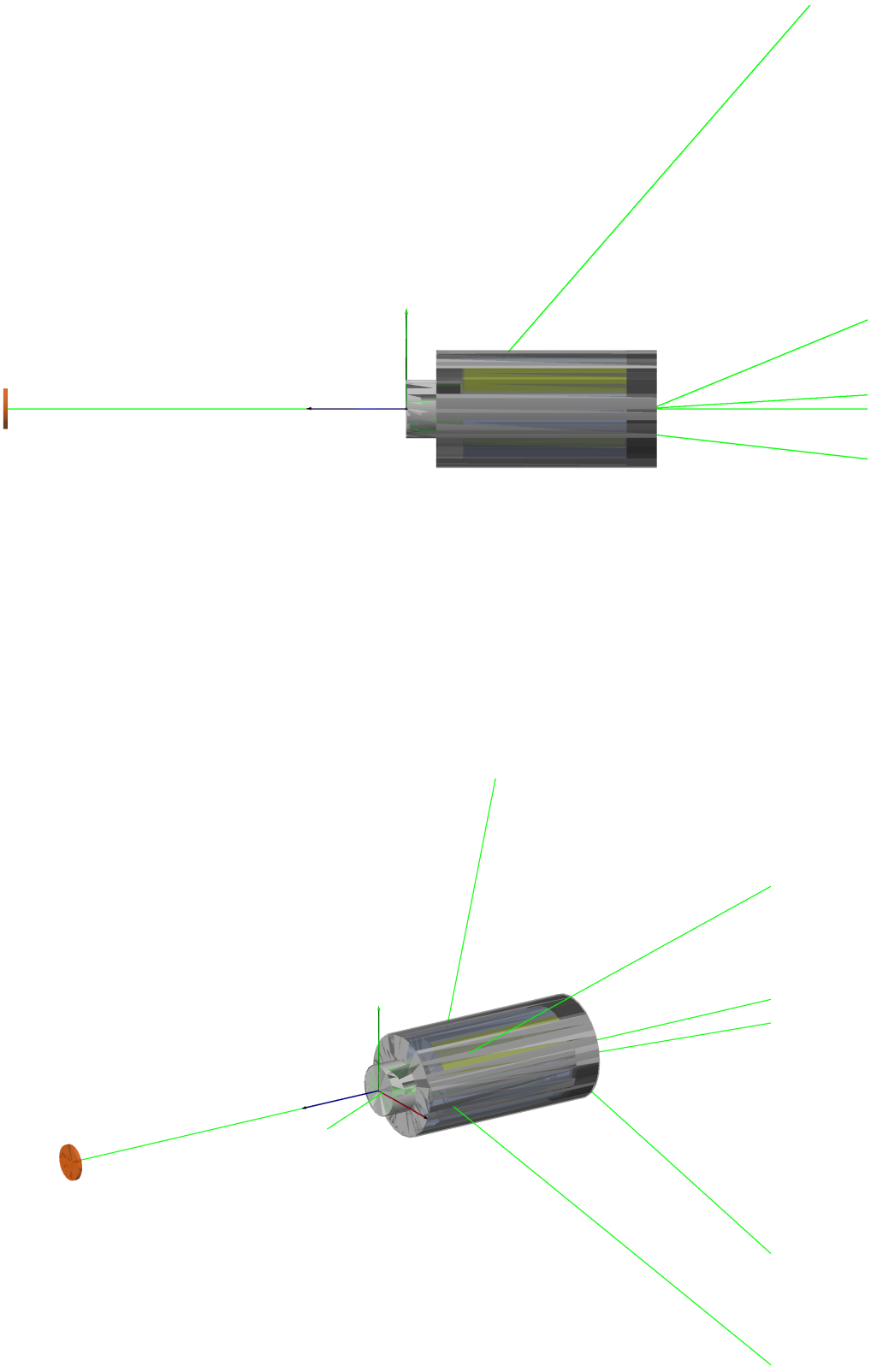


Figure 1: Geant4/OpenGL event displays for ten 661.64 keV gammas from the source puck at 200 mm. Top: side view along the detector axis; the source puck appears edge-on at the left and the detector is at the right. Bottom: oblique view showing the puck, the detector body, the crystal, and the tracks. The source is focused toward the detector only for these renders; the production source is isotropic.

4 Simulation method

4.1 Production matrix and physics

The production matrix is three distances times three energies, with 1×10^7 incident gammas per point. The runs use the `G4DCSmonitor` binary built from this repository, the `GPS` source mode, and the physics list in `PhysicsList.cc`: `G4EmStandardPhysics_option4` for electromagnetic transport, with hadronic HP, thermal-neutron, ion, and gamma-nuclear components included for the full simulation. The gamma efficiency is controlled by the electromagnetic physics and the geometry; the hadronic components do not contribute at these energies.

The source is a point at the puck centre with an isotropic angular distribution. The detector geometry includes the crystal, the detector body, and the magnesium shield. The production macro writes the `hits` tree and the `meta` tree. The `hits` tree stores one row per pulse; the analysis sums all rows with the same event number to recover the total energy deposited in the crystal per event. The analysis uses detector identifier 0 for the CLYC crystal and ignores zero-weight rows.

Table 3: Simulation production matrix and full-energy peak counts. Each point uses 1×10^7 incident gammas. N_{peak} counts events whose total energy deposition lies within $\pm 2\,\%$ of the incident energy. The last column is the relative statistical uncertainty $\sqrt{N_{\mathrm{peak}}}/N_{\mathrm{peak}}$, expressed as a percentage.

d [mm]	E [keV]	N_{inc}	N_{peak}	ϵ_{sim} [%]	rel. stat [%]
200	661.64	1×10^7	843	0.008 43(29)	3.4
200	1173.23	1×10^7	384	0.003 84(20)	5.1
200	1332.51	1×10^7	326	0.003 26(18)	5.5
300	661.64	1×10^7	385	0.003 85(20)	5.1
300	1173.23	1×10^7	166	0.001 66(13)	7.8
300	1332.51	1×10^7	148	0.001 48(12)	8.2
400	661.64	1×10^7	205	0.002 05(14)	7.0
400	1173.23	1×10^7	106	0.001 06(10)	9.7
400	1332.51	1×10^7	71	0.000 71(8)	11.9

4.2 Efficiency definitions and analysis

The simulated full-energy peak efficiency is

$$\epsilon_{\mathrm{sim}} = \frac{N_{\mathrm{peak}}}{N_{\mathrm{inc}}} \times 100, \tag{6}$$

where $N_{\mathrm{inc}} = 1 \times 10^7$ and N_{peak} is the number of events with

$$(1 - 0.02)E \leq E_{\mathrm{dep}} \leq (1 + 0.02)E. \tag{7}$$

The $\pm 2\,\%$ window captures the full-energy peak and excludes the single-escape peak at $E - 511\,\mathrm{keV}$ and the double-escape peak at $E - 1022\,\mathrm{keV}$. The simulation has no detector resolution, so the full-energy peak is a delta function; the window is wide enough to contain it and narrow enough to exclude the escape peaks. The statistical uncertainty is

$$\sigma_{\mathrm{sim}} = \frac{\sqrt{N_{\mathrm{peak}}}}{N_{\mathrm{inc}}} \times 100. \tag{8}$$

The measured efficiency is given by Eq. (1); its statistical uncertainty is dominated by the net peak area,

$$\sigma_{\mathrm{meas}} = \frac{\epsilon_{\mathrm{meas}}}{\sqrt{N_{\mathrm{peak}}}}. \tag{9}$$

The simulation-to-measurement ratio is

$$R = \frac{\epsilon_{\mathrm{sim}}}{\epsilon_{\mathrm{meas}}}, \qquad \sigma_R = R\sqrt{\left(\frac{\sigma_{\mathrm{sim}}}{\epsilon_{\mathrm{sim}}}\right)^2 + \left(\frac{\sigma_{\mathrm{meas}}}{\epsilon_{\mathrm{meas}}}\right)^2}. \tag{10}$$

The pull is

$$\text{pull} = \frac{\epsilon_{\text{sim}} - \epsilon_{\text{meas}}}{\sqrt{\sigma_{\text{sim}}^2 + \sigma_{\text{meas}}^2}}. \tag{11}$$

5 Results

5.1 Efficiency versus distance

Figure 2 shows the simulated and measured full-energy peak efficiencies as functions of distance. The simulation points are filled markers connected by lines; the measurement points are open markers. The vertical axis is logarithmic because the efficiency falls by an order of magnitude between 200 mm and 400 mm. The 661.64 keV curves overlap within the marker size at 200 mm and separate by up to 9 % at the larger distances. The 1173.23 keV and 1332.51 keV simulation curves lie above the measurement at every distance, with the largest separation at 200 mm.

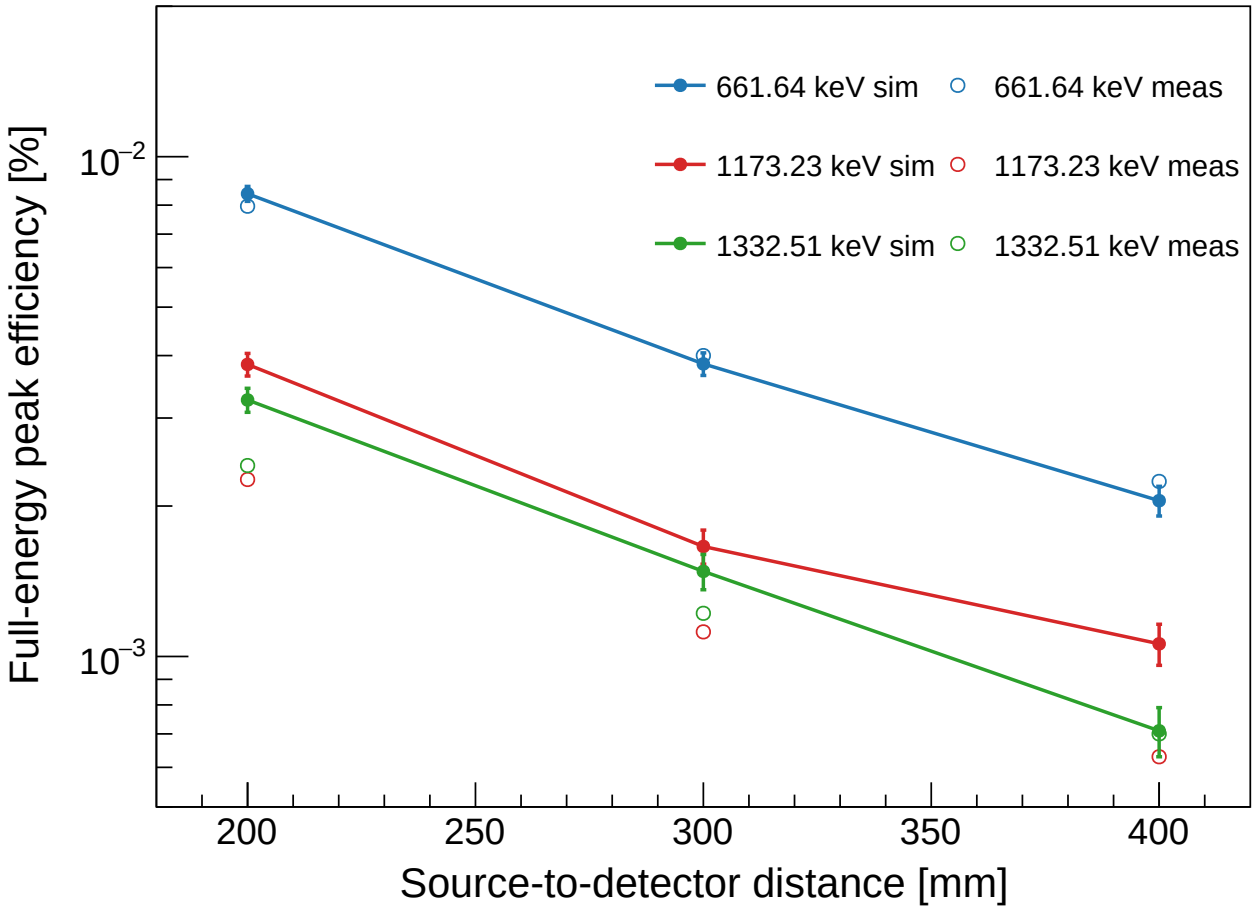


Figure 2: Full-energy peak efficiency versus source-to-detector distance for the three gamma lines. Filled markers and solid lines are the simulation; open markers are the measurement. The vertical axis is logarithmic. The simulation uses 1×10^7 incident gammas per point; the error bars are the statistical uncertainties of Eqs. (8) and (9).

5.2 Ratio and energy dependence

Figure 3 shows $R = \epsilon_{\text{sim}}/\epsilon_{\text{meas}}$. The shaded band is $\pm 10\,\%$ around $R = 1$, and the dashed line is $R = 1$. The 661.64 keV points lie inside the band at all three distances. The 1173.23 keV points lie above the band at all three distances; the ratios are 1.699, 1.482, and 1.683. The 1332.51 keV points are 1.353 at 200 mm, 1.213 at 300 mm, and 1.014 at 400 mm; the first two are above the band, while the last is inside it.

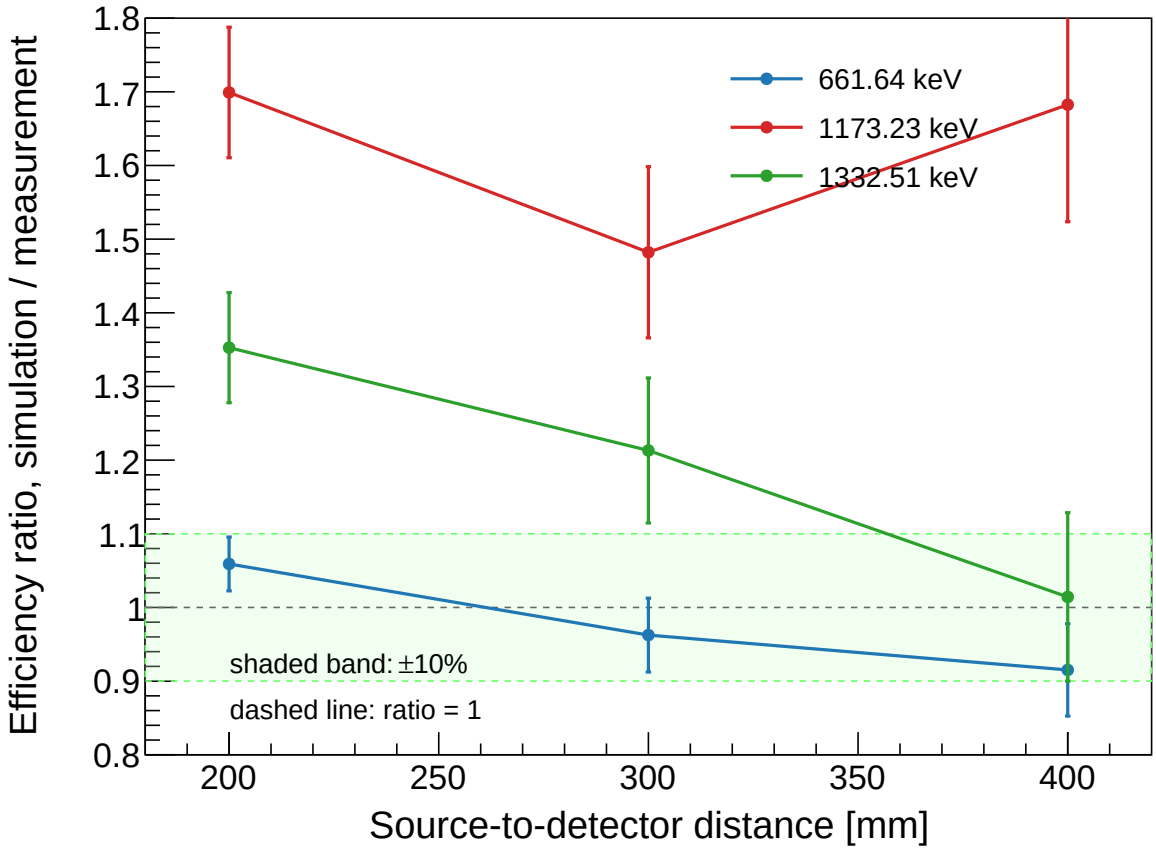


Figure 3: Simulation-to-measurement efficiency ratio versus distance. The shaded band is ±10 % around $R = 1$; the dashed line is $R = 1$. The error bars follow Eq. (10). The 661.64 keV points are consistent with unity; the 1173.23 keV and 1332.51 keV points are high, except for the 400 mm 1332.51 keV point.

Table 4 collects the numerical comparison. The pull column uses Eq. (11). The 661.64 keV pulls are 1.6, −0.8, and −1.4; none is significant at the 2σ level. The 1173.23 keV pulls are 7.9, 4.2, and 4.3; all are significant. The 1332.51 keV pulls are 4.7, 2.2, and 0.1; the 200 mm and 300 mm points are significant, while the 400 mm point is not.

Table 4: Comparison of simulated and measured full-energy peak efficiencies. The measured values are the corrected CSV values from Eq. (3). The ratio and pull follow Eqs. (10) and (11). The simulation uncertainty is statistical only; the measured uncertainty is the Poisson error on the net peak area.

d [mm]	E [keV]	ϵ_{sim} [%]	ϵ_{meas} [%]	R	pull
200	661.64	0.008 43(29)	0.00796	1.059	1.6
200	1173.23	0.003 84(20)	0.00226	1.699	7.9
200	1332.51	0.003 26(18)	0.00241	1.353	4.7
300	661.64	0.003 85(20)	0.00400	0.963	-0.8
300	1173.23	0.001 66(13)	0.00112	1.482	4.2
300	1332.51	0.001 48(12)	0.00122	1.213	2.2
400	661.64	0.002 05(14)	0.00224	0.915	-1.4
400	1173.23	0.001 06(10)	0.00063	1.683	4.3
400	1332.51	0.000 71(8)	0.00070	1.014	0.1

The measured 1173.23 keV/1332.51 keV ratio is 0.94 at 200 mm, 0.92 at 300 mm, and 0.90 at 400 mm. The CSV reference column gives 1.21; the simulation gives 1.18, 1.12, and 1.49. The measured ratio is therefore inverted: the 1332.51 keV peak is larger than the 1173.23 keV peak at every distance, while the reference and the simulation both put the 1173.23 keV peak above the 1332.51 keV peak. The 400 mm simulation ratio of 1.49 is higher than the reference ratio because the 400 mm 1332.51 keV point has only $N_{\mathrm{peak}} = 71$ and an 11.9 % statistical uncertainty; the ratio

is correspondingly imprecise.

5.3 Spectra at 200 mm

Figure 4 shows the simulated total-energy-deposition spectra at 200 mm for the three energies. The histograms are normalised to 1×10^7 incident gammas and to the bin width in keV, so the vertical axis is events per 1×10^7 incident gammas per keV. The dashed vertical lines mark the ±2 % window and the central line marks the incident energy. The 661.64 keV spectrum has a clear full-energy peak at 661.64 keV. The 1173.23 keV and 1332.51 keV spectra have smaller full-energy peaks on a Compton continuum; the peaks are still visible above the continuum, and the ±2 % window contains them.

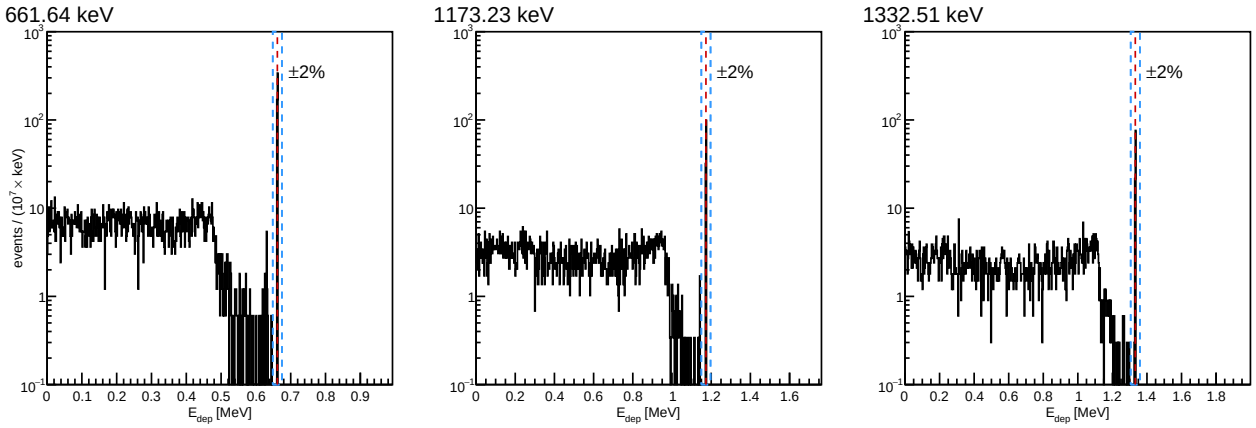


Figure 4: Simulated total energy deposition spectra at 200 mm for 661.64 keV, 1173.23 keV, and 1332.51 keV. The histograms are normalised to 1×10^7 incident gammas per keV. The dashed vertical lines mark the ±2 % window; the central line marks the incident energy. The full-energy peaks are the sharp structures at the incident energies; the lower continuum is Compton scattering in the crystal and the surrounding material.

5.4 No-puck control

The no-puck control uses the same geometry and source position as the 200 mm 661.64 keV point, but omits the epoxy puck. The control gives $\epsilon_{\mathrm{sim}} = 0.008\,76(30)\,\%$, compared with $0.008\,43(29)\,\%$ with the puck. The ratio is 1.100. The fractional change is

$$\frac{\epsilon_{\text{with puck}} - \epsilon_{\text{no puck}}}{\epsilon_{\text{no puck}}} = -3.8(48)\,\%. \tag{12}$$

The puck therefore changes the 661.64 keV efficiency by less than 10 % at the 1σ level. The puck attenuation at 1173.23 keV and 1332.51 keV is smaller than at 661.64 keV, so the puck cannot explain the 1.5–1.7 factor at 1173.23 keV or the 1.2–1.35 factor at 1332.51 keV. The control supports the conclusion that the discrepancy is in the detector response or the source model, not in the source encapsulation.

6 Discussion

6.1 What the 661.64 keV agreement shows

The 661.64 keV ratios are 1.059, 0.963, and 0.915. The corresponding pulls are 1.6, −0.8, and −1.4. The simulation and measurement therefore agree within the combined statistical uncertainties at all three distances. This agreement tests several ingredients at once: the crystal diameter and length, the 2 mm crystal recess, the source-to-detector distance, the ±2 % analysis window, and the conversion from the CSV efficiency to the corrected measured efficiency. A 10 % error in the crystal radius would change the efficiency by about 20 %; a 10 % error in the source distance would change

it by about 20 % at 200 mm and less at 400 mm. Both would be larger than the observed 661.64 keV differences.

The agreement does not validate the absolute efficiency at the 1 % level. The simulation/measurement ratio is 1.059 at 200 mm, 0.963 at 300 mm, and 0.915 at 400 mm. The pattern is not monotonic; it is consistent with the 3–7 % statistical errors of the simulation and the 0.5 % statistical errors of the measurement, plus the systematic uncertainties that are not yet quantified. The 661.64 keV result is a 10 %-level validation, not a precision calibration.

6.2 Why 1173.23 keV and 1332.51 keV disagree

The 1173.23 keV simulation is high by 48–70 %; the 1332.51 keV simulation is high by 1–35 %. The discrepancy is energy dependent: it is small at 661.64 keV, large at 1173.23 keV, and intermediate at 1332.51 keV. A geometric error would scale all three energies similarly, because the geometric acceptance is independent of energy. The 661.64 keV agreement therefore argues against a geometry error as the primary cause.

The most plausible source-model difference is that the simulation generates one monoenergetic gamma per event, while ^{60}Co emits 1173.23 keV and 1332.51 keV gammas in cascade. In the measurement, the two gammas can deposit energy in the crystal in the same event. True coincidence summing can add counts to a full-energy peak (summing in) or remove counts from it (summing out), depending on the source–detector geometry and the surrounding material. The simulation does not include the cascade, so it cannot reproduce either effect. The measured 1173.23 keV/1332.51 keV ratio is inverted relative to the reference and the simulation; cascade summing is a plausible explanation for an energy-dependent change of this sign and size. The present data do not prove this explanation, because the CSV contains net peak areas rather than spectra and the simulation has no coincidence model. A ^{60}Co cascade source in Geant4, with both gammas emitted in the same event, is the direct test.

A second source-model difference is the detector response. The simulation records total energy deposition and applies a ±2 % window. The measurement fits a full-energy peak with a finite resolution and a background. If the 1173.23 keV and 1332.51 keV peaks sit on a larger Compton background than the 661.64 keV peak, the peak-fit area can be biased by the background model. The measured spectra are not in the CSV, so this effect cannot be separated from cascade summing with the present data. A consistent peak-fit procedure applied to the measured spectra, and a simulation with detector resolution and a realistic background, are needed.

A third possibility is an error in the source activity or the emission probability for ^{60}Co. The measured 1173.23 keV and 1332.51 keV efficiencies use the same current activity (11.31×10^6 Bq) and nearly the same emission probabilities (99.97 % and 99.99 %). A common activity error would shift both efficiencies by the same factor and would not invert their ratio. An error in the ratio of the two emission probabilities would have to be about 30 % to explain the measured 0.90 ratio, which is much larger than the quoted values. The activity explanation is therefore less likely than the cascade or peak-fit explanations, but it is not excluded without an independent activity measurement.

The 200 mm measured points are lower than the 300 mm and 400 mm trends. The 661.64 keV 200 mm point is 5.6 % below the simulation, while the 1173.23 keV and 1332.51 keV 200 mm points are 41 % and 26 % below. The close geometry increases the solid angle and the probability of cascade summing, so the larger discrepancy at 200 mm is qualitatively consistent with the cascade hypothesis. A source-distance error of a few millimetres would change the geometric acceptance by a few percent, not by 40 %; the 661.64 keV point at the same distance is consistent with the simulation, so a common distance error is unlikely.

6.3 Intrinsic efficiency

Dividing the full-energy peak efficiency by the geometric acceptance of Eq. (4) removes the leading distance dependence. The 661.64 keV intrinsic efficiency is 0.083–0.094 for both the simulation and the measurement, with no significant distance trend. The 1173.23 keV measured intrinsic efficiency

is 0.024–0.026, compared with 0.039–0.044 in the simulation. The 1332.51 keV measured intrinsic efficiency is 0.025–0.029, compared with 0.029–0.035 in the simulation. The 661.64 keV agreement and the 1173.23 keV disagreement are therefore properties of the detector response, not of the distance scaling. This is the strongest argument that the geometric acceptance and the source placement are not the primary problem.

6.4 Limitations

The comparison has several limitations. The simulation statistical uncertainties are 3–12 %; the 400 mm 1332.51 keV point has only 71 peak events. The measured statistical uncertainties are 0.25–0.5 %, but the systematic uncertainties of the activity, the distance, the detector geometry, and the peak fitting are not included. The simulation has no detector resolution, no scintillation light, no light-collection efficiency, and no electronics threshold. The measured efficiency is a full-energy peak efficiency from a fit; the simulated efficiency is a total-energy-deposition window. The two definitions are close but not identical. The CSV contains net peak areas rather than spectra, so the peak-fit method cannot be checked. The ^{60}Co cascade is not simulated. The reference column of the CSV is not the measured efficiency and is not used as a denominator; it is used only for the energy-dependence comparison. These limitations do not affect the 661.64 keV conclusion, but they limit the interpretation of the 1173.23 keV and 1332.51 keV discrepancies.

7 Conclusion and next steps

The CLYC simulation reproduces the measured 661.64 keV full-energy peak efficiency within 9 % at 200 mm, 300 mm, and 400 mm. The agreement validates the detector geometry, the 2 mm crystal recess, the source distance, the ±2 % analysis window, and the correction of the CSV efficiency by a factor of 1000. The simulation overestimates the 1173.23 keV efficiency by 48–70 % and the 1332.51 keV efficiency by 1–35 %; the 1173.23 keV discrepancy is significant at 4–8σ. The measured 1173.23 keV/1332.51 keV ratio is 0.90–0.94, while the reference and the simulation give 1.21 and 1.12–1.49. The no-puck control changes the 661.64 keV efficiency by 3.8(48) %, so the epoxy puck is not the cause.

The next steps are:

1. Simulate ^{60}Co as a cascade source, with both the 1173.23 keV and 1332.51 keV gammas emitted in the same event, and compare the coincidence-summed full-energy peaks with the measurement.

2. Add a detector response model with energy resolution, a light-collection efficiency, and an electronics threshold, and fit the simulated spectra with the same peak model used for the measured spectra.

3. Obtain the measured spectra and repeat the peak fit with a consistent background model; check whether the 1173.23 keV and 1332.51 keV net peak areas change.

4. Measure the source activity and the source-to-detector distance independently, and propagate their uncertainties into the efficiency.

5. Extend the matrix to more distances and energies, and add a full systematic uncertainty budget before using the simulation for a calibration.

A Reproducing the efficiency scan

The production scan is driven by `scripts/run_clyc_eff_gamma.sh`. The default production mode uses 1×10^7 events per point and the nine-point matrix:

```
cmake --build build -j
scripts/run_clyc_eff_gamma.sh --production
```

The driver writes the macros, logs, and ROOT files to `build/validation/clyc_eff/`. The no-puck control is

```
scripts/run_clyc_eff_gamma.sh --nev 10000000 --nopuck \
  --distances 200 --energies 661.64
```

The efficiency analysis is `scripts/analyze_clyc_eff.C`; one point is analysed with

```
root -l -b -q \
  'scripts/analyze_clyc_eff.C("build/validation/clyc_eff/\
clyc_eff_200mm_661p64keV.root",661.64,10000000,0.00796)'
```

and the same call is repeated for the other eight points with the corresponding energy and corrected CSV efficiency. The report figures are generated by

```
root -l -b -q report/eff_clyc/scripts/make_eff_figs.C
```

The event displays are generated by

```
bash report/eff_clyc/scripts/make_vis_figs.sh
```

The display macros use ten events and a focused source for visibility; they do not change the production macros or the efficiency results.

B Data and code locations

- Measurement: `/home/turko46/data/CLYC_validation/effDist_clyc1.csv`.
- Production outputs: `build/validation/clyc_eff/`.
- Efficiency analysis: `scripts/analyze_clyc_eff.C`.
- Production driver: `scripts/run_clyc_eff_gamma.sh`.
- Production templates: `macros/validation/clyc_eff_gamma_template.mac` and `macros/validation/clyc_eff_gamma_nopuck_template.mac`.
- Detector geometry: `src/GeometryCLYC.cc` and `src/DetectorConstruction.cc`.
- Source placement: `src/PrimaryGeneratorAction.cc`.
- Physics list: `src/PhysicsList.cc`.
- Report figures: `report/eff_clyc/figures/`.
- Figure scripts: `report/eff_clyc/scripts/`.
- Event-display macros: `report/eff_clyc/vis/`.